Colter Snyder

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SUMMARY

Software engineer with two years of professional development in Python, Rust, and C++ on Linux, building scientific simulation software where results must be correct and fast. Built the team's performance-benchmarking and visualization tool, ports numerical models across languages without changing their results, and has taught data structures and algorithms in C++ to 120 students. M.S. in Computer Science (3.97 GPA).

EXPERIENCE

Innovative Scientific Solutions, Inc.

Software Engineer II — *Dec 2024 – Present*

- Spearheaded an internal Python performance-benchmarking tool (a CLI, then a GUI front end) that runs any combination of the platform's 13 effect calculations and charts the results; it exposed a 10x slowdown in our interface to an external library, made the case to leadership with clear graphs, and saved an estimated several weeks of manual testing with hand-crafted JSON inputs.
- Port legacy physics models from Fortran and Python to Rust, preserving their numerical behavior while modernizing the codebase.
- Build Rust and Python (Flask) back-end services with WebSockets and MongoDB, deployed with Docker on Linux, and Angular/TypeScript/Tauri front ends for probabilistic nuclear weapons effects simulation platforms.
- Lead development of individual features on a 4–5 person agile team across the full lifecycle: refine requirements, write object-oriented designs, UML, and design documentation, implement, and verify with automated tests; take part in technical meetings with government customers.
- Review teammates' merge requests under a required separate-reviewer-and-merger process, and maintain code health with Git, GitLab CI/CD, SonarQube, and Python test tooling in ITAR- and CUI-controlled environments.

Tyler Technologies — Lakewood, CO

Software Development Intern — *May 2023 – Aug 2023*

- Diagnosed and repaired a 2,000-test automated suite, restoring about 1,500 tests (75%) to passing and giving the team a reliable foundation for test-driven development.
- Migrated a major legacy VB6 desktop application to object-oriented C# as part of a cloud modernization effort.

Colorado School of Mines — Golden, CO

Postgraduate Research Assistant, Department of Computer Science — *May 2024 – Dec 2024*

- Developed object detection models for semiconductor cleanroom equipment and integrated them, with Python back-end services, into a Unity AR training app, delivering a working demo in collaboration with the University of Arizona (advisor: Dr. Mehmet Belviranli).

Graduate Research & Teaching Assistant — *Jan 2024 – May 2024*

- Researched graph databases for water treatment plant operations on a National Alliance for Water Innovation (NAWI) project (advisor: Dr. Gabe Fierro).

TEACHING

Colorado School of Mines — Golden, CO

Adjunct Professor, Department of Computer Science — *Jan 2023 – Dec 2023*

- Co-taught CSCI 220: Data Structures and Algorithms (C++) to 120 undergraduates, coaching students on algorithm design, complexity analysis, and debugging (Fall 2023).
- Taught a section of CSCI 102: Intro to Computer Science Lab (Python) to 90 undergraduates (Spring 2023).

SKILLS

Languages: Python, C++, Rust, TypeScript, JavaScript, C#, C, Fortran, SQL

Systems & Tools: Linux, Git, GitLab CI/CD, Docker, SonarQube, Flask, WebSockets, MongoDB, Angular, Tauri

Practices: Data Structures & Algorithms, Performance Benchmarking & Visualization, Object-Oriented Design, Test-Driven Development, CI/CD, Agile

Personal Projects: FPGA design (Verilog), 6502 & Z80 assembly programming, CAD/3D printing

EDUCATION

Colorado School of Mines — Golden, CO

Master of Engineering in Nuclear Engineering (in progress) — *Aug 2026 – Expected May 2031*

Master of Science in Computer Science • GPA 3.97 — *Jan 2022 – May 2024*

Bachelor of Science in Computer Science • Magna Cum Laude • GPA 3.77 — *Aug 2019 – Dec 2022*

Relevant coursework: Theory of Computation, Computer Vision, Computer Graphics & Advanced Computer Graphics (C++), Theory of Cryptography

Honors: CACI C-MAPP Scholar, National Merit Scholar, Dean's List, Eagle Scout, Rotary Youth Leadership Award